



Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. Computer Science and Engineering

Course Code & Title: CS8792 Cryptography and Network Security

Name of the Faculty member (s): Dr.R.Venkatesh, ASP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / Transposition Techniques: Columnar Transposition Cipher
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Role play
- **Justification:**
 - Role play exercises give students the opportunity to assume the role of a person or act out a given situation. These roles can be performed by individual students, in pairs, or in groups which can play out a more complex scenario. Role plays engage students in real-life situations. Role Play is the activity where the individual takes the role and act according to the given role.
 - **Transposition Techniques Role-play: Columnar Transposition Cipher** is a kind of game where the students are involved to play the game as per the rule.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - Faculty explained the concept of the problem.
 - Students were asked to write a Plaintext. **Ex. Welcome**
 - Six students were asked to come in front of the classroom and stands as per the key which is integer number **Ex.31425**.
 - Student 1 represent key number 3
 - Student 2 represent key number 1 and so on.
 - Sixth student come to in front of the class and assign a plaintext letter to student 1 as W, student 2 as E, student 3 as L and so on.
 - Finally, Sixth Student write the Cipher Text as per the Key order as below
 - (Student 2, Student 4, Student 1, Student 3, Student 5)
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)



• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	It involves logical thinking and make the students able to think about problem solving approach	Students will be able to understand the problem in real manner than theoretical explanation	Students will be able to provide solution for the problem by understand the problem through this activity	As the students are involved in the activity as team of two, the team work skill improved	Students communication skill will be improved as they are facing audience and communicating the activity	The problem solving skill earned through this activity helps the students in solving various problems

• **Images / Screenshot of the practice:**



Fig 1: Students involvement in Columnar Transposition Cipher

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students said it is quiet easy to understand when compare with other illustrative programs
- Through this activity the students felt easy to understand the problem

❖ **Benefit of the practice:**

- Students understanding of the problem is improved
- Students coding skill through this activity is better

❖ **Challenges faced in implementation:**

- Unable to make more students to involve in this activity

References:

- ❖ <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/CS8493-OS-Roleplay.pdf>
- ❖ https://aminoapps.com/c/attack-on-titan/page/blog/different-types-of-roleplay-lit-team/YoYg_nMTbux63E26M7z6xpRxxkovvoBXPj

Signature of Faculty Member

HOD-CSE